

A Mutual Attention Model for Drug Target Binding Affinity Prediction

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Abstract—Vrious machine learning approaches have been developed for drug-target interaction (DTI) prediction. One class of these approaches, DTBA, is interested in Drug-Target Binding Affinity strength, rather than focusing merely on the presence or absence of interaction. Several machine learning methods have been developed for this purpose. However, almost all depend heavily on the use of increasingly sophisticated inputs to improve their performance. In addition, these methods do not allow any analysis or interpretation due to their black-box characteristic. This work is an attempt to overcome these limitations by taking advantage of the use of attention mechanisms with convolution models. In this paper, we define a new mutual attention based model for DTBA prediction. We represent both compounds and targets by sequences. Our model starts by aligning the drug-target pairs, then a learned masking is performed to retain the most promising regions, of both sequences, and amplify them with a learned factor in such a way to make the learning focus more on them. We evaluate the performance of our method on two benchmark datasets, KIBA and Davis. The results show that our mutual attention approach is very effective. Compared to other well-known approaches, it achieved excellent results regarding the considered performance metrics.

Index Terms—Drug-target binding affinity prediction, deep learning models, attention-based models

1 INTRODUCTION

DETECTING the presence of interactions between a drug and a target is crucial in drug discovery and repositioning. Therefore, various computational methods have been explored to predict potential drug-target interaction, with a minimum cost and error rate. Among these methods, Drug Target Binding Affinity (DTBA) prediction approaches are of the utmost importance. Consequently, an unprecedented interest has been observed in the development of new computational approaches for effective identification of DTBA. This has given rise to various methods of predicting and analyzing drug-target affinities based on the available interaction data [1], [2], [3], [4], [5]. Nowadays, a huge amount of drugs and genomes heterogeneous data is produced and structured in various accessible databases. This has motivated the research community to investigate the development of machine learning (ML) approaches for efficient DTBA prediction. Globally, these methods can be classified in two main classes: “traditional” machine learning approaches and Deep learning (DL) approaches. The most known approaches in the first class are SimBoost [6] and KronRLS [7]. SimBoost used gradient boosting machine and similarity-based information of drug-target pairs along with features extracted from network-based interactions between pairs [6]. KronRLS is based on Regularized Least Squares [7]. It uses 2-dimensional compound similarity-based representations for drugs and Smith-Waterman similarity

representation for targets. These two methods can predict continuous values representing binding affinity scores, and binarized values indicating the presence or absence of interaction. However, the drawback of these methods is that they are either based on similarities or require expert knowledge. Accordingly, a “feature engineering” phase is necessary. Contrasting ML, deep neural networks can automatically extract important features from the raw input data, synthesize and integrate low-level features into high-level features, and capture complicated nonlinear relationships in a dataset [8]. Thus, they excel in learning data hidden patterns. Stimulated by its noteworthy success, DL-based techniques are now being investigated in many complex domains, including bioinformatics such as in genomics studies [9] and quantitative-structure activity relationship studies in drug discovery [10], [11]. DL-based methods for DTBA prediction, usually show best performance compared to other ML-based and conventional algorithms. Recently, some DL-based DTBA prediction approaches have been developed. These employ deep neural networks (DNNs) using different input models for proteins and drugs [12], [13]. Some other approaches developed stacked auto-encoders [14] and deep-belief networks [15]. DL-based algorithms for DTBA prediction, differ from each other regarding two main aspects: The input data representation, and the architecture of the model. Regarding this last point, we can globally classify DL-based methods in three sub-classes: CNN and RNN based architecture, graph-based neural networks (GNN); and finally attention based models. Stacked auto-encoder models based on RNNs and CNNs were among the earliest deep-learning approaches [17]. Their main objective was to define real-valued vector forms for chemical and genomic representation [18]. CNN based methods have also been investigated for learning from drug-target complexes structures [19]. DeepDTA, introduced in [17],

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includes blocks of convolutional layers that learn latent features from drug and target sequences. The authors concluded that CNN could not handle, efficiently, the amino acid order relationship in protein sequences. In an attempt of enhancing DeepDTA performance, the same authors developed two other models investigating the use of various input representations with different models. The first model is WideDTA [20], a CNN-based model that investigated the combination of multiple inputs, by using four information sources: A combination of ligand SMILES and Ligand Maximum Common Substructure (LMCS), for drugs. Whereas for proteins, it used Amino-acid sequences along with Protein Domains and Motifs (PDM). The second model is DeepAffinity [19], it is an approach based on seq2seq model. DeepAffinity uses SMILES for drugs and structural property sequences (SPS) for proteins. These two last approaches, improved slightly the previous results, while making the models very complicated. PADME [21] is another DL-based approach, it has two versions. PADME-ECFP uses the Extended-Connectivity Fingerprint as input representation for drugs; and PADME-GraphConv is based on convolutional graphs [22]. The two versions use the same protein descriptors. Following the same idea of improving the prediction results by using increasingly complicated input representations, other methods investigating other representation forms, have been developed. MDeePred [23] used combined representation of protein features composed of: sequence, structural, evolutionary and physicochemical properties, all incorporated within 2D vectors. A deep CNN uses this representation for DTBA prediction. As can be seen, all the previous approaches have tried to improve performance by diversifying and combining the inputs representation, or even using multiple representations for the same input, which is not very suitable. The Graph-based class appeared with the aim of obtaining better results. In this class, GraphDTA [24] uses a GNN to represent drugs as molecular graphs, that can directly capture atoms bonds. Proteins are represented as sequences. DGraphDTA [25] is an improvement of GraphDTA where the graph representation is extended to proteins. The authors used the Pcons4 tool based on a U-net model to predict the protein contact map, that is used to construct the protein graph, that is later represented by a GNN. Additionally, node features selected around each protein residue are extracted for further processing. These properties include polarity, electrification, aromaticity and so on. Besides, Position-Specific Scoring Matrix (PSSM), is used to represent other features of residue nodes. Finally, all the obtained features are merged and processed together through a GNN to obtain the vector representation of the corresponding protein. The excessive complexity of DGraphDTA is noticeable. It is induced, in one side, by using graph networks, and in another side, by using multiple complex representations for the same input. Although it seems natural to represent molecules and proteins with graphs, it has been indicated in many works that the improvement obtained is not worth the excessively high complexity of the resulting models. For instance, in [21], the authors developed both approaches: with GNN, and using fingerprints, the obtained results for the last method were better than the ones using graphs. Unlike previous methods, few Attention based models have been developed for DTBA prediction.

AttentionDTA [26] is one of these models, that uses CNN architecture with two versions for attention weights computing. DeepPurpose [27] is a library that implements a set of DTBA prediction methods. One of these uses a self-attention based transformer encoder. As in other domains, the advantage of attention based approaches [28] is noticeable in DTBA prediction.

As it can be clearly observed, the previous methods have at least two limitations:

1. Almost all investigated the use of increasingly sophisticated inputs to improve the obtained results: graphs, fingerprints, various representation forms for the same information. . .etc., making the models very complicated.
2. Except the attention based models, they lack any possible interpretation due to their black-box characteristics.

The aim of this paper is to define a new DTBA method that achieves competitive results, while using simple forms of input and offering the possibility of results interpretation. To achieve these objectives, we rely on the strong concept of attention mechanism. Our approach is motivated by the latest progress witnessed by the application of attention based-models in several domains. In fact, drug-target interaction is a problem that needs this mechanism, since different locations of drug and protein don't contribute similarly in the interaction, which suggests that a special attention should be assigned to the interesting areas of both sequences. Moreover, the attention models, that were originally designed for translation, cannot be directly implemented for the drug-target interaction, where there is no source and destination sequence, but rather, there is a pair of sequences having the same role and interacting with each other in some specific locations that need to be focused on. Thus, we define an approach that attempts to get the most out of applying attention mechanisms on convolutional networks.

Contributions. In this paper, we propose a new deep learning model based on mutual attention for DTBA prediction. Our method uses simply, SMILES representation for compounds, and Amino-acid sequence representation for proteins. An alignment vector is computed between the hidden features of the two sequences. The resulting alignments are used to detect the most promising areas from both sequences, and "guide" the learning process towards these regions. The resulting sequences that are mutually "attentive" to each other, are concatenated and finally, a fully connected block concludes the model by predicting the drug-target binding strength. We conducted extensive experiments with the usually used evaluation metrics. The results show that our proposed method is very effective, while maintaining an acceptable level of complexity and giving a possibility for interpretation. The subsequent sections of this paper are organized as follows. Material and Method section presents the datasets that we used. It also introduces the global network architecture, and in more detail, the mutual attention mechanism. The Experiments section describes the experiments conducted. It introduces performance metrics, baseline methods, experimental design, the experimental

TABLE 1
Datasets Summary

Dataset	Number of Drugs	Number of Proteins	Known DTIs
Davis	68	442	30056
KIBA	2111	229	118254

results and presents a case study of interpretation. Finally, the last section concludes the paper.

2 MATERIAL AND METHOD

The drug-target binding affinity prediction problem is formulated as a regression task that requires the model to predict the continuous value representing the binding affinity score. In this section, we introduce our proposed method with the used datasets. We use Amino-acid sequences for proteins and SMILES sequences as a drug representation. Hence, in the data processing phase, the drug and protein sequences were collected by searching their IDs in their respective databases. The drug-target training datasets are then constructed for the next phase. We define a new mutual attention model that allows drug sequence to attend to protein sequence and vice-versa. The most interesting locations for both sequences are “intensified” by a learned factor making the learning process focus effectively on them. The detailed attention model is described in section D.

2.1 Datasets

Only few benchmark datasets have been used to develop DTBA prediction methods. The most used ones for these models are the two large-scale benchmark datasets: Davis dataset, and KIBA (Kinase Inhibitor Bioactivity). These datasets are introduced respectively in [29], [30], and are both large scale biochemical selectivity assays of the kinase inhibitors. Binding affinity scores are computed differently in each dataset. Davis dataset contains selectivity assays of the kinase protein family and the relevant inhibitors with their respective dissociation constant (Kd) values. It contains interactions of 442 proteins and 68 ligands. In Davis dataset, the dissociation constant value is directly provided as a measure of binding affinity. KIBA dataset contains a total of 229 unique proteins and 2111 unique drugs. KIBA scores represent continuous values of the binding affinity that was calculated utilizing Kd, Ki, and IC50 scores. The higher KIBA score indicates a lower binding affinity. Table 1 summarizes the statistics for these two benchmark datasets. KronRLS [7], SimBoost [6] and DeepDTA [17] are among the most known methods using these datasets.

2.2 Drug and Target Representation

Our model uses SMILES as drug representation, and amino-acid protein sequences for target representation. The Simplified Molecular-Input Line-Entry System (SMILES) is a specification in the form of a line notation that uses printable characters for describing the structure of chemical elements: molecules and reactions. Our choice for SMILES as drug representation is motivated by the recent works in drug-target binding affinity prediction, that showed their appropriateness for various prediction models [21]. SMILES is a true language,

although with a small vocabulary size (atom and bond symbols) and only a few grammar rules. In our model, SMILES are represented by a one-hot encoding. Protein sequences are encoded similarly. Both SMILES and protein sequences have variable lengths, we opted for a maximal length of 1200 for proteins and 100 for drugs. The sequences that are shorter than the maximum length are 0-post-padded, while longer sequences are truncated. The maximum length choice is guided by the length distribution of both sequences. We guaranteed that the chosen maximum lengths cover 95% of both proteins and compounds in the datasets.

2.3 Model Architecture

In this section, we present the overall architecture of the proposed model. The attention model is described in detail in the next section. The drug-target binding affinity prediction problem is framed as a regression problem aiming to predict continuous measures representing binding affinity scores of a drug-target pair. In this work, we develop a deep learning model using convolutional neural networks and defining a new attention mechanism. First, a one-hot encoding is applied to each sequence. Then, we use embedding layers to represent sequences’ characters with high-dimensional (128-dimensional) dense vectors. Two 1-D convolutional layers are used to extract hidden features and capture the presence of discriminative features from the sequences. The first convolutional layer has 32 filters each one having a size 4, and the second layer uses 64 filters of the same size. We did not want to make the network more sophisticated by making it deeper, adding filters, or trying other kind of layers because our aim was to focus on the attention mechanism and exhibit the efficacy of the developed mutual attention approach. The obtained feature maps of drug and target are scrutinized by a mutual attention mechanism resulting in new drug and target features that are “attentive” to each other, and that are concatenated and submitted to a fully connected bloc to perform the prediction. The activation function used for the fully connected layer is Rectified Linear Units (ReLU) that have been widely used in deep learning studies. The output layer was activated with the LeakyRelu () function. The whole neural network model was implemented with Keras [31] and Tensorflow [32]. To avoid overfitting, we used Dropout, L2-Regularization, and Batch Normalization techniques [33]. The Mean Squared Error is used as a loss function and Adam [34] as optimizer. The learning was completed with 150 epochs. To have additional control on the learning rate, we used a Callback for initializing the learning rate and automatically tuning it during the training and reduce its value based on the monitored parameter (the loss function), the initial learning rate value was set to 0.01, and the minimum value was set to 0.0001. This allowed a quick loss decreasing in the beginning, and then, when the loss value stopped decreasing, the learning rate value was automatically reduced. The network architecture is shown in Fig. 1.

2.4 Mutual Attention Model

This section presents our mutual attention model. We note H_d and H_p : The drug and target hidden features obtained from the previous convolutional layer. $H_d \in \mathbb{R}^{C \times N}$ and $H_p \in \mathbb{R}^{C \times M}$ where c is the channels number, N is the number

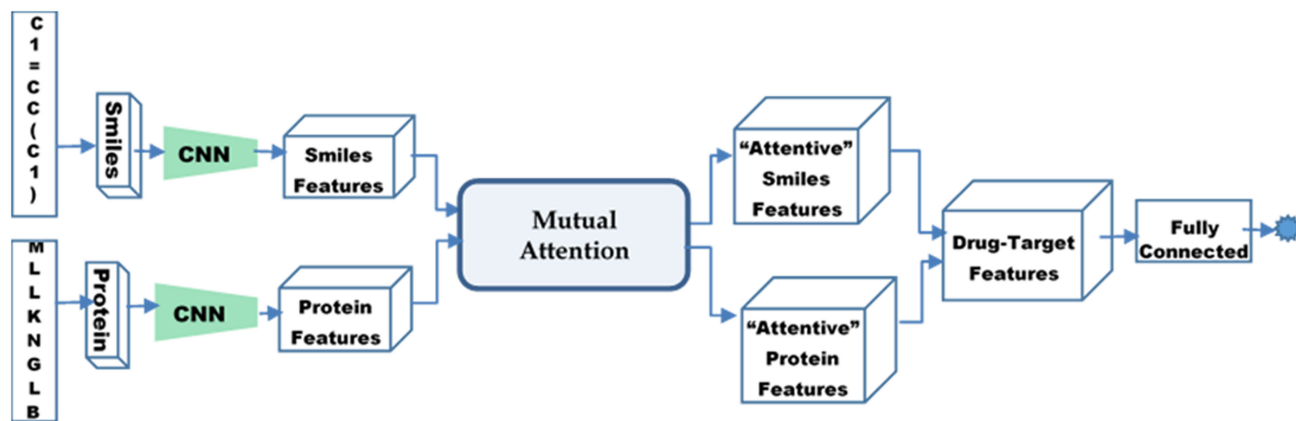


Fig. 1. A mutual attention based model for DTBA prediction.

of feature locations in the drug and M is the number of feature locations in the protein. First, an alignment is computed between H_d and H_p . The obtained alignment scores are used to compute both drug attention weights (DAW) and protein attention weights (PAW). Drug attention weights represent the attention that should be given by the drug to each protein location, and inversely, protein attention weights represent the attention that should be given by the protein to each drug location. The drug-protein pair attention weights (DPAW) are obtained by adding PAW and DAW.

The intuition behind computing PAW and DAW separately then adding them is that the habitually used Softmax () function is computed based on one axis, by default the last one. But in our case, the aim is to detect the mutual promising areas in both drug and protein. This is obtained by computing the Softmax () separately along the drug axis and the protein axis. We Add the scores because we aim to give importance to the locations that are interesting from both sequences. We tried various other combination methods, as multiplication, maximum, and average, the best results were obtained by using the addition. Afterwards, two learned variables, that we called: Theta and Gamma, are used to accentuate more on the potential interesting areas of the sequences. Theta is used as a learned threshold to mask all the values of DPAW which are less than it, we call the obtained result MaskedDPAW. A 1x1 convolutional layer is used as a projection technique, to project H_d and H_p into two feature spaces H_{dt} and H_{pt} . This way allows to perform a dimensionality reduction, by decreasing the number of feature maps while retaining their salient features. H_{dt} and H_{pt} are both multiplied by MaskedDPAW,

resulting respectively in H_{dm} and H_{pm} . Gamma is used as a learned factor by which H_{dm} and H_{pm} are multiplied to make the learning process focus adaptively more on them. Finally, the obtained results are added to H_d and H_p to constitute the mutual attention outputs. We set the initial value of Gamma to 0 to make the learning adaptively progress from the easier to the more complicated tasks. The equations are presented in Fig. 2, a graphical representation is presented in Fig. 3.

3. EXPERIMENTS AND RESULTS

In this section, we describe the experiments we conducted for our proposed model, which is a Mutual attention based deep learning model for drug-target binding affinity prediction method. We implemented our model in Python with Keras and Tensorflow [31], [32]. We analyzed the effectiveness of the various techniques used in this research. Finally, we made a comparison of our approach against other similar methods.

3.1 Evaluation Metrics

Various performance metrics have been employed to assess the performance of prediction approaches and to compare classification models. For the models that specify the problem as regression, the natural metrics are various forms of errors: mean square error, mean average error, and others. To assess the performance of our model, we have selected two metrics, the first being the usual mean square error MSE. We used another metric that is suitable for our study: The Concordance Index [35]. In the subsequent, we will introduce briefly this metric and explain its appropriateness for our work.

Mean Squared Error: MSE. Mean Square Error (MSE) is commonly used in regression tasks to measure how close the line obtained by connecting the predicted values to the actual data points. The formula below defines the MSE, where P denotes the vector of predicted values, Y denotes the vector of the actual outputs, and n is the number of samples. The smaller the MSE, the better the performance of the regressor.

$$MSE = \frac{1}{n} \sum_{i=1}^n (P_i - Y_i)^2$$

- $A = \text{Align}(H_p, H_d) = H_d * H_p^T$ (1)
- $DAW = \text{Drug Attention Weights} = \text{Softmax}(A, \text{axis}=1)$ (2)
- $PAW = \text{Protein Attention Weights} = \text{Softmax}(A, \text{axis}=0)$ (3)
- $DPAW = DAW + PAW$ (4)
- $\text{MaskedDPAW} = \text{Masking}(\text{ThresholdedReLU}(\text{theta} = \text{Theta})(DPAW))$ (5)
- $H_{pt} = \text{Convolutional-Projection}(H_p)$ (6)
- $H_{dt} = \text{Convolutional-Projection}(H_d)$ (7)
- $H_{dm} = \text{MaskedDPAW} * H_{dt}$ (8)
- $H_{pm} = \text{MaskedDPAW} * H_{pt}$ (9)
- $HP' = \text{Gamma} * H_{dm} + HP$ (10)
- $HD' = \text{Gamma} * H_{pm} + HD$ (11)

Fig. 2. Mutual attention equations.

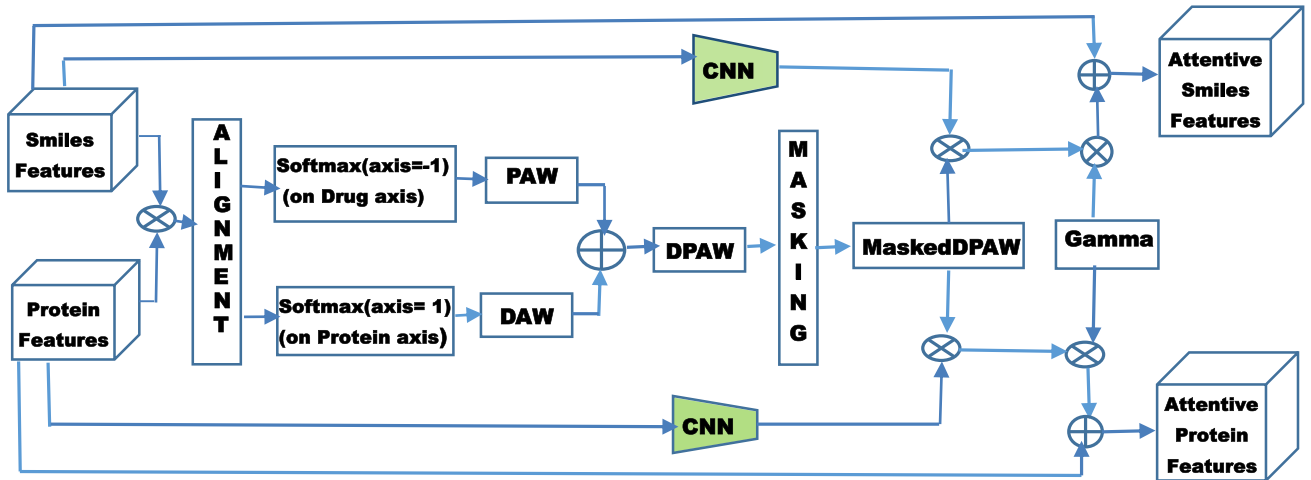


Fig. 3. Mutual attention model.

Concordance Index: CI. The concordance index or C-index (or CI) can be considered as a generalization of the area under the ROC curve (AUC) that is usually used in binary classification. It represents the global assessment of the model discrimination power: this is the model's ability to correctly provide a reliable ranking of the drug-target pairs. The concordance index of a set of data considers each data pair and calculates the probability that the order of the predicted label values is the same as the order of the true label values. As shown in Fig. 4, the prediction f_i of the higher affinity y_i is greater than the prediction f_j of the smaller affinity value y_j [7]. When the main objective is the prediction of the labels relative order rather than their values, the CI measure is a suitable performance metric. For example, in DTBA investigations, it is desirable to have an estimation of the interaction likelihood of a given drug (or target) with all the targets and to rank accordingly the targets (or drugs) [7]. The CI is the proportion of the concordant pairs over the total number of pairs. Its formula is given in Fig. 4.

Where the function h is defined by:

$$h(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0.5 & \text{if } x = 0 \\ 0 & \text{if } x < 0 \end{cases},$$

Z is a normalization constant representing the number of pairs with different labels, and $h(u)$ is the step function. The values of the CI range between 0.5 and 1.0 [7]. Ties in the predictions are counted as half concordant pairs. Random predictions have an average concordance index of 0.5. Hence, a model without any learning skills has a mean concordance index of 0.5. The maximal value of CI is 1.0, it corresponds to the perfect order of predictions regardless of the quality of the prediction accuracy, as the concordance index is interested exclusively in the order of the predictions, it does not give any insight on the quality of the

$$CI = \frac{1}{Z} \sum_{y_i > y_j} h(f_i - f_j)$$

Fig. 4. Concordance index.

predictions themselves. Thus, for instance, it will be equal to the maximal value 1 if the order is correct even if the predictions are inaccurate.

3.2 Experimental Design

To assess the performance of our method, we used cross-validation (CV), which is the conventional approach in recent researches. This allows a more comprehensive coverage of the complete datasets, which provides a more accurate evaluation of the model's performance. As it is the usual tradition in previous related works, the number of folds was set to 5. For every dataset, and for every CV splitting, we evaluated the prediction performance. Deep learning models' performance depends highly on various hyperparameters like learning rate, number and size of filters, number of layers, and other parameters. As explained before, for the learning rate, we opted for a dynamic tuning by using a Keras Callback that starts with an initial value of the learning rate, and reduces it dynamically during the training when the monitored loss value is stagnant, meaning that the model is not improving. Additionally, we considered two hyperparameters the number and the size of the filters for each layer in the convolutional block. We tuned manually the hyperparameters based on validation results, the hyperparameters combination having the best validation results was selected. Purposely, we did not focus more on the selection of hyperparameters since our main objective was rather to show the effectiveness of our mutual attention model. Overfitting is a common problematic behavior of deep learning models that have been targeted by many researchers. This has led to the apparition of various techniques, among which dropout, regularization and batch-normalization are the most used methods. Dropout consists of hiding randomly a given proportion of nodes during the training process. This allows a better generalization of the model by preventing it to be too dependent on training data. In our model, we used dropout with various percentages in almost all the layers. We included a Batch-normalization in many positions of our model since it is usually recommended with convolutional layers. We also penalized the loss function with regularization L2-norm [36]. Finally, we updated the weights using the Adam optimizer with a

penalized loss to give a generalized prediction for the model. Embedding vector values are randomly initialized by the glorot initializer, which imposes normal distribution of weights and variance of output following variance of input. The model was constructed using Keras and Tensorflow.

3.3 Baseline Methods

To assess the performance of our work, we compared it with seven state-of-the-art methods, namely: SimBoost [6], KronRLS [7], DeepDTA [17], WideDTA [20], MdeePred [23], GraphDTA [24] and AttentionDTA [26]

1. **SIMBOOST** [3] is a gradient boosting machine-based method that constructs feature for each drug, each target, and each drug–target pair. With each drug–target pair, SimBoost associates a feature vector. SimBoost uses three networks: drug–drug similarity network, target–target similarity network, and drug–target binding network. In the latter case, a node can either be a drug or a target and the drug nodes and target nodes are connected to each other via the binding affinity value. Latent vectors from matrix factorization are also included in this network. A supervised learning method named gradient boosting regression trees [37] is used for DTI prediction.
2. **KronRLS** [7] is based on Regularized Least Squares Models (RLS) and it can predict both binary classes and continuous binding values. Given a set $\{d_i\}$ of drugs and a set $\{t_j\}$ of targets, the goal of KronRLS is to learn a prediction function $f(x)$ for all possible drug–target pairs $x \in \{d_i \times t_j\}$. An objective function computing the dissimilarity between two pairs has to be learned. Thus, the problem of learning the prediction function f is formulated as finding a minimizer of the corresponding objective function.
3. **DeepDTA** [17] is a deep-learning model based on Convolutional Neural Network (CNN) architecture. It includes two distinct stacked CNN blocks, one for protein sequences and one for SMILES that learn separately latent features of drug and target, followed by pooling layers reducing the size. DeepDTA performance was assessed on the two datasets: Kiba and Davis. It was designed for continuous values prediction, but was also assessed for binary classification.
4. **WideDTA** [20] is a CNN model that attempts to improve DeepDTA results by using four input representations. WideDTA represents SMILES and Amino Acid sequences as sets of words instead of complete sequences. Word lengths were 3 for proteins and 8 for drugs, respectively. Four models with the same architecture are used to extract features from each of the input information. The obtained features from each model are then concatenated and fed to a fully connected block to predict the binding affinity.
5. **MDeePred** [23] is another approach that aims at defining new input features for protein. It uses combined representations of protein features composed of: sequence, structural, evolutionary and physico-chemical properties, all incorporated within 2D

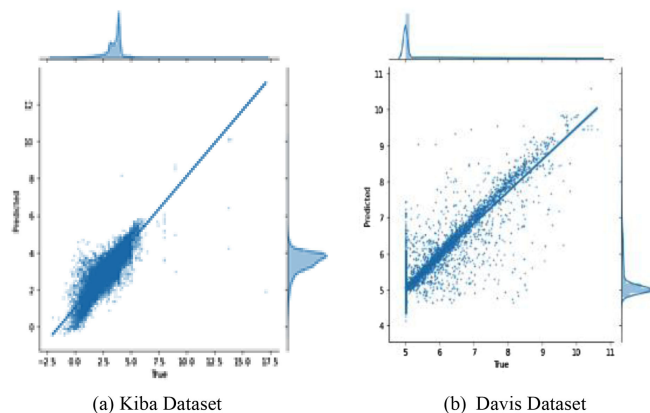


Fig. 5. Predicted-actual plots.

vectors. A deep neural network uses this representation to predict the drug–target binding affinity.

6. **GraphDTA** [24] uses graph neural networks to predict drug–target affinity. In this model, drugs are represented as molecular graphs, that can directly capture the bonds among atoms, and proteins as sequences. The model latent space is statistically analysed to show that the graph neural network can automatically focus on well-defined chemical features and correlate between hidden node activations and domain-specific drug annotations.
7. **ATTENTIONDTA** [26] uses convolutional neural networks with attention to predict the binding affinity score of a drug–target pair. Two versions have been defined: Attention1DTA and Attention2DTA. The same formulas are used in both versions except that for the first version the same weights are used to compute the attention scores for the drug and target, while for the second version, dissimilar weight matrices are used.

3.4 Results

This section presents our experimental results. We have assessed our model for the considered datasets: Davis and Kiba. In the experiments reported in this paper, each dataset was split into two sets, train set and test set using 5-fold cross-validation. Our method is a deep learning model that uses mutual attention mechanism to predict the binding strength of a drug–target pair. Figs. 5a and 5b show the graphs representing the predicted vs Actual binding values of Kiba and Davis datasets. The graphs show clearly that the model succeeded to predict the actual binding scores with a very small error. The Tables 2 and 3 summarize our results and those of baseline methods, in terms of Mean Square Error (MSE) and Concordance Index (CI) metrics. The average values scores obtained by performing 5-fold cross-validation are reported. The results for the baseline methods were taken from the experiments reported in the literature [6], [7], [17], [20], [23], [24], [26]. Our model showed the best performances in both metrics for both datasets. For Kiba dataset (Table 2), our method is followed by GraphDTA for both metrics. For Davis dataset (Table 3), our model achieved the best performances followed by GraphDTA and Attention1DTA, for CI score; and Attention1DTA for MSE metric. these results are all the more valuable since they were

TABLE 2
CI and MSE Scores for KIBA Dataset

	CI	MSE
KronRLS	0.782	0.411
SimBoost	0.836	0.222
DeepDTA	0.863	0.194
WideDTA	0.875	0.179
Attention1DTA	0.882	0.155
Attention2DTA	0.880	0.162
GraphDTA	0.891	0.139
MDeepred	NA	NA
Our Method	0.8984	0.1372

TABLE 3
CI and MSE Scores for DAVIS Dataset

	CI	MSE
KronRLS	0.871	0.379
SimBoost	0.872	0.282
DeepDTA	0.878	0.261
WideDTA	0.886	0.262
Attention1DTA	0.893	0.216
Attention2DTA	0.886	0.222
GraphDTA	0.893	0.229
MDeepred	0.88	0.26
Our Method	0.9051	0.1593

obtained by a model having an acceptable complexity, compared to other models, in particular to graph-based models.

3.5 Case Study Interpretation

One of the objectives of our work was to avoid the black-box characteristic by allowing, to some extent, an analysis of the obtained results. The importance of such a feature is two-fold. First, it can be used for already known drug-target interactions to confirm prior knowledge. Additionally, it is very useful for unidentified drug-target interactions and drug discovery, as having an idea of interesting locations for both drug and target can guide laboratory experiments to focus exclusively on these regions' components, which implies a reduction in costs. To show how our method could be used for this purpose, we present a case study of a sample taken from the Davis dataset. It is related to the compound-protein pair having the sequences given in Fig. 6a and 6b. Fig. 7 shows the attention weights of drug-target location pairs, obtained for these sequences. The drug is represented on the X axis, while the protein is represented on the Y axis, at a scale of 1/10; given its great length. Based on their attention weights heatmap, we can see that some regions contribute actively to the interaction. Those are the locations with the higher attention weights in the heatmap, they are colored in green. Whereas other locations seem inactive, since they have low attention weights. For instance, for the Smiles sequence, the locations around: 14 to 16, from 34 to 37, and to a lesser degree, locations from 60 to 68, seems interesting. For the protein sequence, locations from 85 to 95; 260 to 270; 280 to 290; and from 375 to 385 show potential interest. Having this information along with the predicted binding score, a

MGDEKDSWKVKTLDLQEQKKRRKEQEEAEIKRLKNSDDDRSKRDSLEEGELRDH
CMEITIRNSPY RREDSMEDRGEDDSLAIKPPQMSWKKEVHHRKDEKRKEK-
CRH HSHSAEGGKHARVKEREHER RKRHREEQDKARWEWERQKRREAREHSR-
RER D RLEQLERKRERERKMREQQKEQREQKERERR AEERRKEREAREVSAHH
RTMRED YSDKVKASHWSRSPRP RPREFELGDGRKPVKEEKMEERDL

LSDLQDISDSEERKTSSAE**SAESGSGSEEE**EEEEEEEEEEEGTSEEEEEEE
EEEEETGSNSEEASQSAEEVSEEESEDEERENENHLLVVPESRFRDSEGESEAE
VGE**TPQSSALTEG**DYVPDSPALLPIELKQELPKYLPALQGCRSVVEFQCLNRREE
GTYGVVYRAKDKKTDEIVALKRLKMEKEKEGFPITSLREINTILKAQHPIVTVREIV
VGSNMDKIYVMNYVEHDLKSLMETMKQFPLPGEVKTLMIQLLRGVKHLHDNWI
LHRDLKTSNLLSHAGILKVGFGLAREYGSPLKAYTPVVVTQWYRAPPELLGAKE
YSTAVDMWSVGCIFGELLTQKPLFPNGNSEIDQINKVKELGT PSEKIWPGESELPVV
KKMTFSEHPYNNLRKRFGALLSDQGFDMNKFITYFPGRISAEDGLKHEYFRETP
PIDPSMFPTWPAKSEQQRVKRGTSRPPPEGGLGYSQLGD DDLKETGFHLTTTNGQ
ASAAAGPGFSLKF

(a)

C[C@@]12[C@@H]([C@@H])(C[C@@H])(O1)**N3C4=C**C=CC=C4C5=C6C(=C
7C8=**C=C=C8N**2C7=C53)CNC6=O)N(C(C(=O)C9=CC=CC=C9)OC

(b)

Fig. 6 (a) Protein sequence. (b) Drug sequence.

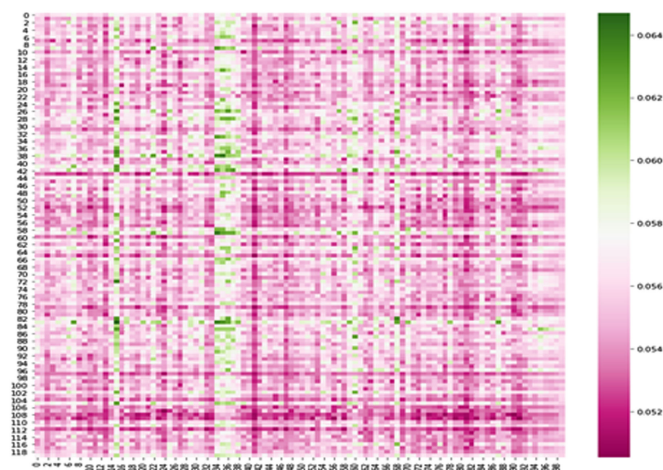


Fig. 7. Attention weights heatmap.

biologist could easily validate it either by his prior knowledge if the interactions are already known; or, employ it by conducting experiments that focus directly on these specific areas, in the opposite case. The underlined parts of Fig. 6 highlight some interesting locations in both drug and target sequences, based on the information provided by the attention weights heatmap. To exhibit more the advantage of our method regarding interpretability, we present a 3D-view of the previous complex. The SMILES sequence above corresponds to *Staurosporine* compound, which is a protein kinase inhibitor that has been used extensively to induce apoptosis (death of cells) in variety of cases, including especially, cancer and tumour cells. The protein sequence is related to the Cyclin-Dependent Kinase 11A. Fig. 8 presents a 3D-view of the structure that contrasts the areas that have been identified as “interesting” by our model, with previously known binding regions. The regions that are coloured in pink corresponds to known binding regions that are also detected by our attention model. The region coloured in blue is identified by our model as having high attention weights but is not a previously known binding region. This example shows clearly the power of our method in both prediction and interpretability.

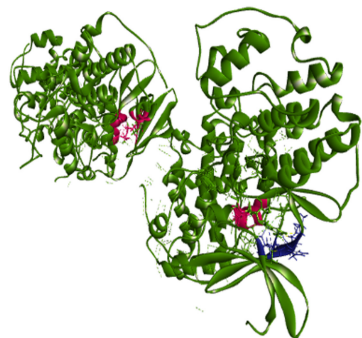


Fig. 8. 3D-view of the complex of staurosporine with cyclin-dependent kinase 11A.

4 CONCLUSION

In this paper, we introduced a new attention-based deep learning approach for DTBA prediction. Our aim was to tackle the drug-target binding affinity problem more efficiently while avoiding black-box and exceedingly complicated models. Our model is based on a mutual attention mechanism and convolutional Neural Networks. Our work is motivated by the latest successful applications of attention based-models in several domains. In fact, drug-target interaction is a problem that needs this mechanism since different drug and protein locations don't participate similarly in the interaction, which suggest that a special attention should be assigned to interesting areas of both sequences. However, the attention models that were, originally, defined for translation and other similar tasks in natural language processing, cannot be directly implemented for drug-target interaction, since in an interaction, there is not a source and destination sequence, but rather a pair of sequences having the same role, and that need to attend to each other. Thus, we defined a mutual attention model resulting in a pair of sequences where each sequence is 'aware' of the most interesting parts of the second sequence. The resulting alignments are used to "guide" the learning process towards the most promising regions of both sequences, and amplify them with a learned factor in such a way to make the learning focus more on them. We compared the performance of the proposed model with some state of the art methods in DTBA prediction. We perform our experiments on the commonly used datasets: Davis and KIBA. Our results are very convincing and outperform the other methods across both datasets and for both evaluation metrics. Furthermore, our method showed a suitable level of interpretation that was not possible for the other methods. Our results indicated also that the use of more sophisticated forms for the input is not sufficient to describe drug-protein interactions, rather than that, focusing on interesting regions in both drug and protein could be the key idea. We may try further enrichments of our system by defining other forms of attention mechanisms, investigating more drug-target binding interpretation, and exploring the use of our mutual attention model for binding sites prediction

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